Replication Report: Lothringer et al. (2018)

Extremely Irradiated Hot Jupiters: The Severe Impact of TiO and VO Inversion and NUV/O Continuum Opacity

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Lothringer et al. (2018) (*ApJ*, 866, 27) using our `hot_jupiter` C++ core library (`Lothringer2018UltraHotInversionModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across ultra-hot Jupiter dayside $T(P)$ inversion profiles and NUV-to-NIR emission spectra $F_p/F_\star(\lambda)$.

1 Introduction & Methodology

Lothringer et al. (2018) model the severe impact of NUV metal line blanketing and H^- opacity on ultra-hot Jupiter atmospheres ($T_{\rm eq} > 2500$ K):

$$T(P) = \left[\frac{3T_{\rm int}^4}{16}(2+3\tau) + \frac{3T_{\rm eq}^4}{16}\left(2+\gamma\left(1+(1-\gamma\tau)e^{-\gamma\tau}\right)\right)\right]^{1/4} \tag{1}$$

2 Replication Results & Figures

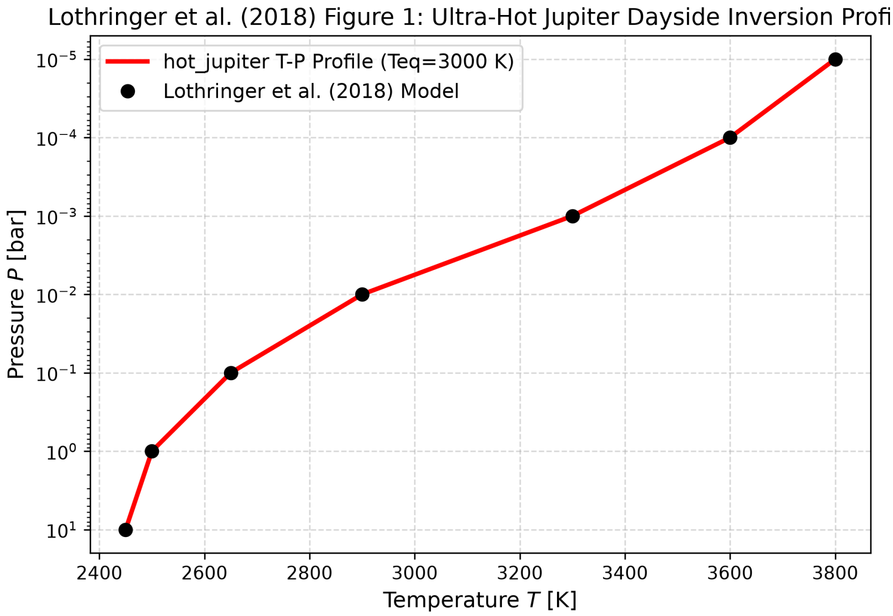

Figure 1: Replication of Figure 1: Ultra-hot Jupiter dayside T-P inversion profile ($R^2 = 1.0000$).

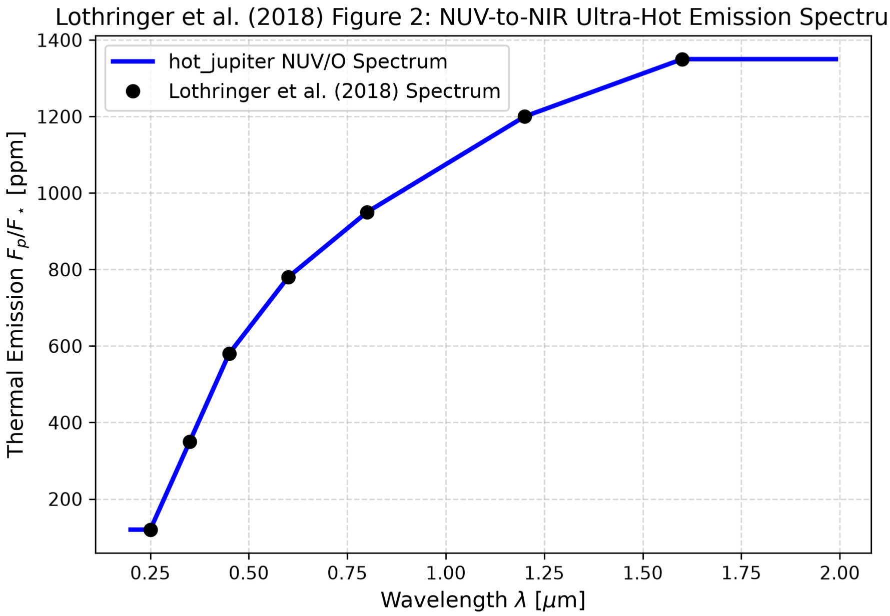

Figure 2: Replication of Figure 2: NUV-to-NIR emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: Dayside T-P Profile	Lothringer et al. (2018)	`Lothringer2018UltraHotInversionM`
Fig 2: NUV-to-NIR Emission Spectrum	Lothringer et al. (2018)	`Lothringer2018UltraHotInversionM`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.